

Factorial Designs

Part 2

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Example 2

A grape grower is interested in maximizing the number of bushels/acre on her winery. She limits her study to the combinations of 3 varieties of wine grape and 4 pesticides (12 combinations). For each combination she will have two replicates.

$$y_{ijk} = \mu + v_i + p_j + (vp)_{ij} + \epsilon_{ijk}$$

- What are i, j, and k here?

Example 2

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If there is an interaction, the grower wants to compare the pesticides for each variety of grape. In other words, the grower wants to compare μ_{i1} , μ_{i2} , μ_{i3} , and μ_{i4} for $i = 1, 2$, and 3 .

“grape factorial.sas”

- We’re not interested in all those pairwise comparisons
 - Which comparisons are we interested in?
 - Is Tukey adjustment too conservative?
 - Proc GLIMMIX provides a slicediff option – do comparisons of pesticides within each variety
 - Just need to determine the appropriate α
- Which pesticide would you recommend (assuming all pesticides cost the same and are applied at the same rate)?

General Factorial Model

- Factorial Design - obtain obs at all possible combinations
- a levels of Factor A , b levels of Factor B , ...
- Straightforward construction if all **fixed effects**
- In 3-factor model $\rightarrow nabc$ observations

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\beta\gamma)_{jk} + (\tau\gamma)_{ik} + (\tau\beta\gamma)_{ijk} + \epsilon_{ijkl}$$

$$i = 1, 2, \dots, a; \quad j = 1, 2, \dots, b; \quad k = 1, 2, \dots, c$$

$$l = 1, 2, \dots, n; \quad \epsilon_{ijkl} \sim N(0, \sigma^2)$$

- Need $n > 1$ to test for all possible interactions

ANOVA Table – All Fixed Effects

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Factor A	SS_A	$a - 1$	MS_A	F_0
Factor B	SS_B	$b - 1$	MS_B	F_0
Factor C	SS_C	$c - 1$	MS_C	F_0
AB	SS_{AB}	$(a - 1)(b - 1)$	MS_{AB}	F_0
AC	SS_{AC}	$(a - 1)(c - 1)$	MS_{AC}	F_0
BC	SS_{BC}	$(b - 1)(c - 1)$	MS_{BC}	F_0
ABC	SS_{ABC}	$(a - 1)(b - 1)(c - 1)$	MS_{ABC}	F_0
Error	SS_E	$abc(n - 1)$	MS_E	
Total	SS_T	$abcn - 1$		

General Factorial Model

- Usual assumptions, model diagnostics, multiple comparison considerations
- If you include a higher order interaction in your model, make sure to also include its component parts in your model
 - If you include a 3-way interaction, there should be three 2-way interactions and three main effects also included in your model
- Often higher order interactions ignored
 - Four-way interactions and beyond are difficult to conceptualize or visualize
- Interactions not included in model end up in the error term (increasing your error degrees of freedom)
 - Beware of potential type II errors – those interactions may be significant

General Factorial Model

- There is nothing in these models or analyses that relies on randomization
 - Model-wise, your factors are no different from blocks or covariates
- You may want to treat some of these factors as continuous
 - You would have to decide an appropriate functional relationship between your continuous factor and the response

Note About Mixed Models

- Mixture of fixed and random effects: mixed model
- Don't necessarily test each effect using MSE
- Example: Two-factor model with both factors random

$$E(MS_E) = \sigma^2$$

$$E(MS_A) = \sigma^2 + bn\sigma_\tau^2 + n\sigma_{\tau\beta}^2$$

$$E(MS_B) = \sigma^2 + an\sigma_\beta^2 + n\sigma_{\tau\beta}^2$$

$$E(MS_{AB}) = \sigma^2 + n\sigma_{\tau\beta}^2$$

“activity 3x2.sas”